

Chapter 2: Application Layer

Numerical Problems

Problem 1: DNS and HTTP Delay

A user clicks on a link in a Web browser. The IP address corresponding to the URL is not available in the local cache, so a DNS lookup is required.

Suppose that n DNS servers are contacted before the host obtains the IP address. The successive DNS queries incur round-trip times

$$RTT_1, RTT_2, \dots, RTT_n.$$

Let RTT_0 be the round-trip time between the local host and the Web server. The requested Web page contains exactly one object consisting of a small amount of HTML text.

Assume that non-persistent HTTP is used and that the object's transmission time is negligible.

Determine the total time from clicking the link until the object is received.

Problem 2: HTTP Response Time

Consider the scenario in Problem 1. Suppose that the HTML file references eight additional very small objects located on the same server.

Assume that:

- the transmission times of all objects are negligible;
- a new TCP connection must be established before the first HTTP request is sent;
- all required DNS lookups have already been completed;
- for the parallel case, assume that up to six objects are downloaded simultaneously. Once a connection becomes available, it is used to download one of the remaining objects.

Determine the elapsed time for retrieving the base HTML file and all eight referenced objects for each of the following cases:

1. Non-persistent HTTP with no parallel TCP connections.
2. Non-persistent HTTP with the browser configured to use six parallel TCP connections.
3. Persistent HTTP without pipelining.

Problem 3: Web Caching

Consider an institutional network connected to the Internet through a 15 Mbps access link. The average object size is

1,000,000 bits,

and the average request rate from the institution's browsers to origin servers is

16 requests/second.

The average Internet delay, measured from forwarding an HTTP request at the Internet-side router until the corresponding response is received, is

3 seconds.

Model the total average response time as the sum of the average access delay and the average Internet delay.

For the average access delay, use

$$\frac{\Delta}{1 - \Delta b},$$

where Δ is the average time required to send an object over the access link and b is the arrival rate of objects to that link.

Assume that the delay for retrieving an object from the local cache is negligible.

For the cache case, calculate the average response time by considering that a fraction equal to the cache hit rate experiences negligible delay, while a fraction equal to the cache miss rate experiences the corresponding access and Internet delays.

1. Find the total average response time when no cache is used.
2. Suppose that a cache is installed in the institutional LAN and the cache miss rate is 0.4. Find the new total average response time.

Problem 4: Parallel and Persistent HTTP

Consider a short 10-meter bidirectional link with a transmission rate of

150 bits/sec.

Data packets have length

100,000 bits,

while control packets, such as ACK or handshaking packets, have length

200 bits.

Suppose that HTTP is used to download objects of size

100 Kbits,

and the initially downloaded object references 10 additional objects from the same sender.

Assume that when N parallel TCP connections are used, each connection receives an equal share of the link bandwidth. For this problem, take

$$N = 10.$$

Also assume that persistent HTTP is used without pipelining.

1. Calculate and compare the total download time using non-persistent HTTP without parallel connections and non-persistent HTTP with 10 parallel TCP connections. State which approach results in the smaller download time.
2. Calculate the total download time using persistent HTTP without pipelining. Compare the calculated download time with that of non-persistent HTTP and state which approach results in the smaller download time.

Problem 5: HTTP/2 Frame Interleaving

A Web page is transmitted using HTTP/2. The page consists of one video clip and five images.

The video clip is divided into

2000 frames,

and each image consists of

3 frames.

Assume that transmitting one frame requires one frame transmission time.

Assume that, for the round-robin transmission case, the transmission order begins with the video clip, followed by Image 1, Image 2, Image 3, Image 4, and Image 5.

1. If all video frames are transmitted before any image frames, determine the number of frame transmission times required before all five images have been completely transmitted.
2. Suppose that the frames are transmitted in an interleaved manner such that one frame from each available object is transmitted in round-robin order. Determine the number of frame transmission times required before all five images have been completely transmitted.

Problem 6: HTTP/2 Prioritization

Consider the Web page described in Problem 5. HTTP/2 prioritization is now used.

All images have higher priority than the video clip. Furthermore, the first image has higher priority than the second, the second image has higher priority than the third, and so on.

Assume that a higher-priority object is completely transmitted before transmission of a lower-priority object begins.

Determine the number of frame transmission times required until the second image has been completely transmitted.